

MATH 345 Skeleton Notes

Unit 6: Constraints and SVD

Section 6.2: Lagrange multipliers and parallel gradients

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Note: Lagrange multipliers as parallel gradients The Lagrange multiplier condition $\nabla f(a) = \lambda \nabla g(a)$ means that, at a constrained optimum, the objective gradient is parallel to the constraint gradient. Equivalently, along the allowed tangent directions to the constraint, there is no first-order increase or decrease.

This uses four earlier ideas: dot products, orthogonality, level sets, and gradients.

We now discuss the method of Lagrange multipliers, which is a method of finding potential local minima and local maxima of a function on a *hypersurface* S in $\mathbb{R}^n$ given as the level set of a function, i.e. when $S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\}$, for some differentiable scalar-valued function g and some $k \in \mathbb{R}$.

Definition: Critical Points Consider a surface of the form $S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\}$, where g is a differentiable scalar-valued function with $\nabla g(\mathbf{x}) \neq 0$ for all $\mathbf{x} \in S$. If a point $\mathbf{a} \in S$ is in the interior of the domain of a differentiable function f , then we say that $\mathbf{a}$ is a

constrained critical point of f on S if there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a}).$$

Equivalently, define the

Lagrangian function

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - k).$$

Then $\mathbf{a}$ is a constrained critical point exactly when there exists $\lambda \in \mathbb{R}$ such that

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{a}, \lambda) = \mathbf{0}, \quad \frac{\partial \mathcal{L}}{\partial \lambda}(\mathbf{a}, \lambda) = 0.$$

The *method of Lagrange multipliers* is a way of finding optima of a function f along a hypersurface S given as above. The method consists of finding all critical points of the function f on S (of which there are usually only finitely many). This consists of finding the solutions to the system of vector equations $g(\mathbf{x}) = k$ and $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$. The next theorem shows that any local extremum of f on S must occur at a critical point, so that the method of Lagrange multipliers will find these extrema.

Theorem Consider a surface

$$S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\},$$

and assume ∇g is nonzero on S . If a point $\mathbf{a} \in S$ is a local extremum of a differentiable function f on S , and $\mathbf{a}$ is an interior point of the domain of f , then $\mathbf{a}$ is a constrained critical point of f on S .

In particular, if S is closed and bounded and f is continuous on S , then absolute extrema of f on S exist. Under the same differentiability and nonzero-gradient hypotheses, every absolute extremum of f on S is among the constrained critical points.

Proof The idea is quite simple when sketched geometrically. In order for $\mathbf{a}$ to be a local extremum for f on S , it must be true that for any unit vector $\mathbf{v}$ in the tangent plane to S at a point $\mathbf{a}$, the directional derivative $D_{\mathbf{v}} f(\mathbf{a})$ must be equal to 0 (or else travelling along S in the direction

of $\mathbf{v}$ or $-\mathbf{v}$ will lead the function to increase or decrease, which contradicts the fact that $\mathbf{a}$ is a local extrema). Recalling Definition def-10-1-directional-derivative, it follows that

$$\nabla f(\mathbf{a}) \cdot \mathbf{v} = 0$$

for any vector $\mathbf{v}$ in the tangent plane to S at $\mathbf{a}$, which can *only* be true if $\nabla f(\mathbf{a})$ is normal to the tangent plane to the surface S at $\mathbf{a}$. When S is equal to the level set of a function g , a normal vector to the tangent plane at $\mathbf{a}$ is given by $\nabla g(\mathbf{a})$. So if $\mathbf{a}$ is a local extremum of f on S , then $\nabla f(\mathbf{a})$ and $\nabla g(\mathbf{a})$ must be multiples of one another, i.e. it must be true that

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a})$$

for some $\lambda \in \mathbb{R}$, i.e. so that $\mathbf{a}$ is a constrained critical point of f on S . The final statement follows from the extreme value theorem: if S is closed and bounded and f is continuous on S , then absolute extrema exist, and any absolute extremum is also a local extremum on S . The first part of the theorem then shows that such a point must be a constrained critical point.

Activity

Find the absolute extrema of $f(x, y) = 3x + y$ subject to the constraint $x^2 + y^2 = 10$.

Write $g(x, y) = x^2 + y^2$. Then we are finding extrema of f on the surface

$$S = \{(x, y) : g(x, y) = 10\}.$$

We calculate that

$$\nabla f(x, y) = (3, 1) \quad \text{and} \quad \nabla g(x, y) = (2x, 2y).$$

So the constrained critical points of f on S are the points $(x, y) \in S$ for which there exists λ such that

$$(3, 1) = \lambda(2x, 2y),$$

i.e. the points that satisfy the three equations

$$\begin{aligned}x^2 + y^2 &= 10 \\2\lambda x &= 3 \\2\lambda y &= 1.\end{aligned}$$

If the second and third equation holds, then λ cannot be zero, and we can divide both sides

of the equation by λ to obtain that $x = 3/2\lambda$ and $y = 1/2\lambda$. Substituting this into the first equation, we find that

$$(3/2\lambda)^2 + (1/2\lambda)^2 = 10,$$

which simplifies to $1/4 = \lambda^2$, so either $\lambda = 1/2$ or $\lambda = -1/2$. If $\lambda = 1/2$, then $x = 3$ and $y = 1$, and if $\lambda = -1/2$, then $x = -3$ and $y = -1$. So the two points $(3, 1)$ and $(-3, -1)$ are the two constrained critical points of f on S . Since S is a circle, and is closed and bounded, the absolute maximum and minimum occur among these two points. We have

$$f(3, 1) = 10 \quad \text{and} \quad f(-3, -1) = -10,$$

so $(3, 1)$ is the absolute maximum of f on S , and $(-3, -1)$ is the absolute minimum of f on S .

Activity

Find the absolute extrema of the function $f(x, y, z) = x^2 + 2y^2 + 3z^2$, subject to the constraint $x^2 + y^2 + z^2 = 1$.

Let $g(x, y, z) = x^2 + y^2 + z^2$. Then $\nabla f(x, y, z) = (2x, 4y, 6z)$ and $\nabla g(x, y, z) = (2x, 2y, 2z)$. A point (x, y, z) is a constrained critical point of f on the sphere $S = \{(x, y, z) : x^2 + y^2 + z^2 = 1\}$ only when there exists $\lambda \in \mathbb{R}$ so that the two equations

$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$ and $x^2 + y^2 + z^2 = 1$ both hold, which expands to the four equations

$$2x = \lambda 2x$$

$$4y = \lambda 2y$$

$$6z = \lambda 2z$$

$$x^2 + y^2 + z^2 = 1.$$

Since $x^2 + y^2 + z^2 = 1$, either $x \neq 0$ (and dividing the first equation by x we obtain $\lambda = 1$), $y \neq 0$ (and dividing the second equation we obtain $\lambda = 2$), or $z \neq 0$ (and dividing the third equation by z gives $\lambda = 3$). When $\lambda = 1$, the

first equation automatically holds, and the second and third equation can only hold when $y = z = 0$. Thus the fourth equation becomes $x^2 = 1$, so $x = \pm 1$. Similarly, when $\lambda = 2$ we have $x = z = 0$ and $y = \pm 1$, and when $\lambda = 3$ we have $x = y = 0$ and $z = \pm 1$. So the function f has six constrained critical points on S , at the points $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$, and $(0, 0, \pm 1)$. We have

$$f(\pm 1, 0, 0) = 1 \quad f(0, \pm 1, 0) = 2 \quad \text{and} \quad f(0, 0, \pm 1) = 3.$$

So the absolute minima of f occur at $(\pm 1, 0, 0)$, and the absolute maxima of f occur at $(0, 0, \pm 1)$.